NATIONAL UNIVERSITY

Third Year Syllabus

Department of Library and Information Science (LIS)

Four-Year B.A. (Honours) Course

Effective from the Session: 2013–2014

NATIONAL UNIVERSITY

Syllabus for Four Year B.A. Honours Course
Subject: Library and Information Science (LIS)
Session: 2013-2014

THIRD YEAR

Paper Code	Paper Title	*Marks*	*Credits*
233801	Information Networking and Resource Sharing	100	4
233803	Database Design and Application	100	4
233805	Documentation	100	4
233807	International and Comparative Librarianship	100	4
233809	Research Methodology and User Studies	100	4
233811	Automation of Library and Information Centres	100	4
233813	Indexing and Abstracting	100	4
233815	Information and Communication	100	4
	Total =	800	32

OBJECTIVES

(1) To give the students an understanding of the basic principles of fundamental laws of Library and Information Science and to enable them to understand and appreciate the functions, purposes and services of Libraries and Information Organizations in this fast changing world of 'Information Age'.

(2) To train the students in the techniques of Information Management and equip them with the latest developments in Information Technology (IT) and its applications in libraries and information centres.

(3) To acquaint the students with the organizations and development of the Universe of Knowledge.

Details Syllabus

Paper Code	233801	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	**Information Networking and Resources Sharing**			

Unit 1 Resource sharing and information networking: definition, origin and gradual development; scope and importance of information networking; fields and objectives of information resource sharing; functions and activities of information networking and resource sharing

Unit 2 Basic agreements of library networking and resource sharing; limitations/barriers; methods of information networking and resource sharing; components of networking; essential factors, tools and logical techniques for networking and resource sharing

Unit 3 Networking configurations: definitions, types, procedures, tools and techniques; network topologies

Unit 4 Application and usefulness of computer technologies in information networking; LAN, WAN and Internet; types of LAN and WAN; network protocols

Unit 5 Introduction to Internet; Internet application and services; tools of Internet; operational problems and Internet communications; Web pages and Web sites

Unit 6 National, regional and international library networking systems in major information institutions around the world; problems and prospects of library networking system with special reference to Bangladesh

Reading list:

Chandel and Veena S. Planning Library Resource Sharing.
Arnold, D.V. The Management of the Information department.
Eyer, J. and Kent, A. Resource Sharing Techniques.
Henely, J. P. Computer Based Library and Information System.
Marlin, E. W. Computers and Information System.
Tonks, P. Computers and System: an introduction for Librarians.

Paper Code	233803	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	**Database Design and Application**			

Unit 1 Basic concepts of data, information, bit, byte, field, and record; evolution, purpose and functions of database

Unit 2 Organization of file: direct access files; sequential access files; data files; program Files: inverted/indexed files; buffer management; file addressing; access methods, physical, sequential, indexed sequential, index random, direct, inverted, hashing; data retrieval, flat, hierarchical and multiple structure; data storage devices, different secondary storage devices

Unit 3 Database design: functional steps in database design; database models, relational DM, hierarchical DM, network DM; data integrity; file systems; access control; data security, back-up and recovery

Unit 4 Introduction to some available database management software: DBMS packages; recent trends in library database management; problems and prospects of database management systems in Bangladesh

Unit 5 Development of database using CDS/ISIS package: CDS/ISIS system overview, hardware requirement, software installation, steps in database development in ISIS environment, designing data structure, data entry worksheet, inverted file and display format, prepared data for computer use, rules for data entry, searching and retrieval logic, sorting and printing. Exporting and importing data, data security, control and recovery, customizing the package (Practical)

Reading list:

Buxton, A. and Hopkinson, A. The CDS/ISIS Handbook.
Everest, G. C. Database Management: Objectives, System Functions and Administration.
Kooth, H. F. and Silberschatz, A. Database System Concepts.
Martin, J. Computer Database Organization.
Wiederhold, G.O. Relational Databases: Concepts, Selection and Implementation

Paper Code	233805	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Documentation**			

Unit 1 Basic concept of document, documentation and information retrieval; meaning and definitions of documents, kind of documents, sources of documents

Unit 2 Documentation: definitions and development, aims, scope and types, active and passive, documentation work and services, distinctions and relationships among documentation, librarianship and information science.

Unit 3 Documentation techniques: preparation of documentation lists, document collection and scanning, content analysis, file organization

Unit 4 Concepts, types, purposes, methods o disseminating current information, criteria for assessment or current information, current awareness service (CAS), definition, purpose characteristics, types and channels, selective dissemination of information (SDI) service, definition, techniques and procedures of providing SDI service, conversion process of CAS into SDI, difference between CAS and SDI, user profile, definition, principle of construction, characteristics, model of user profile, methods of notification.

Unit 5 Documentation and information services in Bangladesh with special reference to BANSDOC, BIDS, ICCDDR,B, AIC, etc.

Reading list:

Guha, B. Documentation and Information retrieval.
Sengupta, B. and Chatterjee, M. Documentation and Information Retrieval.
Lancastester, F. W. Information Retrieval System.
Rowley, J.E. and Turner, C.M.D. The Dissemination of Information.
Rowley, J. Mechanized In-house Information System
Vickery, B.C. Techniques of Information Retrieval.

Paper Code	233807	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**International and Comparative Librarianship**			

Unit 1 Definition of comparative librarianship; comparative librarianship and international librarianship; benefits and goals; comparative librarianship and comparative sciences; comparative method as a scientific method; types of comparative librarianship

Unit 2 Library systems in USA, UK, JAPAN and SAARC countries; public libraries and media centres; academic Libraries; special libraries and information centres; national libraries

Unit 3 The role of national and international organization; role of library associations in the development of libraries in both developed and developing countries

Unit 4 Education for librarianship in developed and developing countries: USA, UK, Pakistan, India and Bangladesh

Unit 5 Library legislations in USA, UK, India, Pakistan and Bangladesh

Unit 6 Interlibrary cooperation and resource sharing in different countries

Unit 7 Application of new technologies in the UK, USA, Japan, India and Bangladesh

Unit 8 Application of information technologies in Bangladesh with comparison to developed and developing countries

Reading list:

Kawatra, P. S. Comparative and International Librarianship.
Kawatra, P. S. Encyclopedia of Library and Information Science Education.
Allen K. Encyclopedia of Library and Information Science.
Whathy, A. International and Comparative Librarianship Group Handbook.
Parker, J. S. Unesco and Library development Planning.
Foskett, D. J. Readers in Comparative Librarianship

Paper Code	233809	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Research Methodology and User Studies**			

Unit 1 Concept of Research: historical growth and development of research, its objectives, types and significance. Research paradigm: positivist, interpretative, action research model, research processes, criteria of good research, qualities and ethics of research, area of research in library and information science; different methods of research

Unit-2 Research problem: concept of research problem, Characteristics of research problem; sources of information, personal experience, literature review, theories, paradigm, Hypothesis, scope of hypothesis, testing of hypothesis, selection of research problems, techniques involved in defining a problem, different steps involved in writing a research proposal.

Unit 3 Research design (RD): Meaning, need, features of a good research design, concepts relating to RD, different types of RD, basic principles of experimental design.

Unit 4 Methods of research: survey, case study, historical documentary research, characteristics, advantages, limitations and techniques, collection of primary data, observation method, interview method, questionnaire method, other methods, their advantages and disadvantages, designing questionnaire and structured interview

Unit 5 Sampling: concepts, characteristics, requirement of a good sample, types of sampling, random and purposive sampling, systematic sampling, cluster, multiple sampling, sampling errors

Unit 6 Data collection techniques: questionnaires, mailed, structured, non-structured, close-ended and open-ended questionnaire; Interviews, structured and non-structured; Observations, structured, non-structured

and participatory; selection of appropriate method in terms of research problems, understanding sources and note taking

Unit 7 Data Processing and Analysis: Editing- central editing, field editing; coding and decoding; tabulation; application of statistics and statistical package in data processing

Unit 8 User studies: definition, importance, types of library user, methods and techniques of user studies

Unit 9 Report writing and evaluation: writing a research report, characteristics of a good research report; designing framework of a research report, drafting report, reviewing and editing, rules for using quotations, footnotes, references and bibliographies, Techniques and criteria for the evaluation of a report

Reading list:

Good, C. and Scates, D.E. Methods of Research.
Kothari, C.R. Research Methodology: methods and techniques.
Kumar, K. Research Methods in Library and Information Science.
Nachimias, C. F. and Nachimias, D. Research Methods in social sciences.
Raj, H. Theory and Practice in Social Research.
Stewart N. B. Operations Resaerch methods: A general approach in primary health care PRICOR.
Zainul Abeden, M.A. A Handbook of Research for the Fellows of M.Phil. and Ph.D. Programmes.

Paper Code	233811	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Automation of Library and Information Centres**			

Unit-1: Library automation: definition and basic functions and component of an automated library system of library automations. Operational functions: Acquisition, cataloguing; processing, serial control, services to the users, circulation function, reference service, collection maintenance, accounting and budgeting, public relations

Unit-2: Automated Acquisition: application of IT, computer aided verification of user frequents, online and offline access, access to publishers databases, ordering, accession and notification to users

Unit-3: Automated processing: locating, creating, editing, machine readable cataloguing, development and maintenance of CD-ROM databases

Unit-4: Circulation functions: introduction to different types of circulation systems (CS) and application of IT, technology/instruments required in a CS; Data item to be considered in CS. Record Keeping, inventory and overdue functions; charging systems; creating and maintaining user profiles; checking materials in and out; reservation systems

Unit-5: Information Systems: access to in-house databases and retrieval issues, CD-ROM search, access through online web-pages, web-sites and remote CD-ROM data bases. Application of IT in different services: CAS, SDI, Preparation of reading list, bibliographies, indexing, etc.; information storage and retrieval issues

Unit-6: Automated support function: Library correspondence, inventory, electronic spreadsheet, database creation and maintenance, management issues in library automation, recent treads and development of library automation in Bangladesh

Reading list:

Kazi Mostak G. H. Tatthya Protishthaner Shayangkriyakaran
Duvel, B. K. and Linda M. Automated Library Systems: a librarian's guide and teaching manual.
Kent, A. Planning for Library Automation.
Ojha, D.C. Computer Application in Library & Information Science.
Rice, J. Introduction to Library Automation.
Rowley, J. Computer for Libraries.

Rowley, J. The Electronic Library.

Paper Code	233813	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	**Indexing and Abstracting**			

Unit-1: Indexing: definition of index, origin and development, importance and types of indexes-author index, alphabetic subject index, classified, cumulative and collective and subject index

Unit-2: Indexing methods: steps in indexing, indexing aids, rules, manuals, operatives system; subject determination, different techniques, citation indexing, Pre co-ordinate indexing, post co-coordinating indexing, chain indexing, POPSI, PRECIS, KWIC, KWOC, etc.; rules of arranging index entries; tools for an indexes

Unit-3: Book indexing: principles techniques and entries, heading, subheading, style and layout

Periodical indexing: techniques and arrangement; newspaper indexing; indexing non-book materials including music, sound recordings, films, etc.; computer-based indexing systems

Unit-4: Concepts of indexing language: vocabulary control; designing a treasures; subject determination

Unit-5: Index evaluation: different methodologies, recall, precision and cost-analysis

Unit-6: Definition, importance, aims and objectives, types of abstracts; qualities of a good abstract; abstract vs. bibliography; abstract vs. index

Unit-7: Methods and procedure of abstracting, international standards of abstracting, evaluation of abstracts; online abstracting systems

Unit-8: Recent trends is indexing and abstracting services; problems and prospects of indexing of abstracting services in Bangladesh

Reading list:

Borka. H. B. C.L. Abstracting Concepts and Methods
Collison, R.L. Indexes and Indexing.
Doxle, L.D. Indexing and Abstracting.
Knight, N. G. The Indexes: training indexing.
Rowley, J.E. Abstracting and Indexing
Bryandt, M. S. Bibliographies, Abstracts and Indexes
Oswald, V. A & Others. Automatic indexing and abstracting of the contents and documents

Paper Code	233815	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	**Information and Communication**			

Unit 1 Nature of Information: Concepts of data, information and knowledge; Data processing; Nature, characteristics and properties of information; Types and parameters of information; Information explosion; Theories of information

Unit 2 Information and Social Change: Changing context of the society; Information as a human resource and its role in social change; Information society; Infoculture; Information science and its linkage with other disciplines

Unit 3 Information Need and Access to Information: Information and its use; Nature of information need/different approaches of information; Information needs covering a wide cross-section of people; Barriers to access to information; Functions of information services in overcoming the difficulties of information users

Unit-4 Information Resource Management (IRM): Definition, objectives and elements of Information Management; Functions of IRM; Techniques of IRM; Role of IT in IRM; Global IT-management; Global challenges of IRM; Ethical and social dimensions of IT; MIS; Information management and policies at national and international levels: national information policy; duties and responsibilities of an information manager

Unit-5 Information Systems: Concept of information systems; Components and types of information systems; Information systems crime; Managing information systems performance and security

Unit-6 Information communication: Information cycle: generation, collection, storage, transmission, and dissemination of Information; Communication: definition, communication process; Functions/role, forms/types, good qualities of communication; Barriers to communication; various models of communication

Unit-7 Media of communication: oral, documentary, audio-visual, mass media, and electronic media of communication; Library as a media of communication; Communication tools and techniques; Internet and its impact on library environment

Unit-8 Telecommunications: concept of telecommunications; telecommunications network model (components); factors affecting telecommunications; role of telecommunications in information systems; problems and prospects of telecommunications in Bangladesh

Reading list:

Foskett, A. C. The subject approach to information.
Gupta, B. In the library and information science horizon
Omar A. M. Business communication: theory and application,
Parida, B. Studies on information systems, services and programs in India and abroad.
Roise, A. M. Information and communication theory
Sharma, C. K. and Arya, S.B. Documentation, information science and automation in libraries.